

Analysis of a Doubly Spread Channel

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1 Theoretical Framework

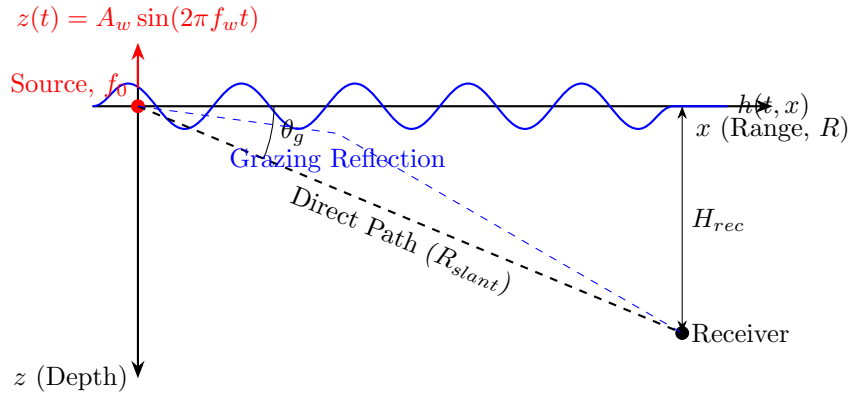
This section establishes the mathematical and physical foundations required to understand how ocean environments distort acoustic signals. By defining the underlying mechanisms of signal spreading, we create a baseline for analyzing specific source geometries.

The theoretical framework defines a **Doubly Spread Channel** [1, 2] as one driven by two coupled mechanisms: **Source Kinematics (Heave)** and **Dynamic Boundary Interaction (Micro-Multipath)**.

2 Source on the surface

A surface vessel represents the most extreme case of acoustic smearing. This section analyzes this geometry to demonstrate the worst-case scenario, where the acoustic source is physically coupled to the dynamic ocean boundary, resulting in maximal delay and Doppler spread.

2.1 Schematic Diagram



2.2 Physical & Mathematical Breakdown

2.2.1 The Grazing Angle

This subsection calculates the geometry of the reflection to prove that for a surface source and distant receiver, the acoustic path reflects at an extremely shallow angle, forcing the direct and surface-scattered paths to overlap almost perfectly.

The grazing angle θ_g is the angle between the direct acoustic path and the horizontal at the point of reflection. It is given by:

$$\theta_g = \arctan\left(\frac{H_{rec} - H_{src}}{R}\right) \quad (1)$$

Because the source is on the surface ($H_{src} \approx 0$) and the range is long ($R \gg H_{rec}$), the grazing angle θ_g approaches 0° . For example, in a setup where the receiver is at 20m depth and the source is 1km away on the surface, $\theta_g \approx 1.15^\circ$.

2.2.2 The Transmitted Signal: Source Kinematics

This subsection models the physical motion of the vessel to prove that its vertical heave inherently transforms the emitted acoustic wave into a Phase/Frequency Modulated (FM) signal before it even propagates into the channel.

A theoretical underwater acoustic source emits a monochromatic pressure wave $p(t)$ at a fundamental frequency f_0 :

$$p(t) = P_0 \cos(2\pi f_0 t) \quad (2)$$

Since a surface vessel is physically coupled to the ocean surface, it undergoes vertical heave motion. We approximate the dominant ocean swell as a harmonic oscillation with amplitude A_w and frequency f_w :

$$z(t) = A_w \sin(2\pi f_w t) \quad (3)$$

This physical motion acts as a time-varying phase shift on the emitted acoustic wave. The transmitted signal $p_{tx}(t)$ becomes a Phase/Frequency Modulated (FM) signal:

$$p_{tx}(t) = P_0 \cos(2\pi f_0 t + \beta \sin(2\pi f_w t)) \quad (4)$$

The severity of the phase modulation is defined by β , which relates the physical heave amplitude to the acoustic wavelength ($\lambda = c/f_0$) and the grazing angle (θ_g):

$$\beta = \frac{2\pi f_0}{c} A_w \cos \theta_g \quad (5)$$

The instantaneous phase $\Phi(t)$ of the transmitted pressure wave is:

$$\Phi(t) = 2\pi f_0 t + \beta \sin(2\pi f_w t) \quad (6)$$

Differentiating the phase $\Phi(t)$ with respect to time yields the instantaneous frequency $f_i(t)$, defined as $f(t) = \frac{1}{2\pi} \frac{d\Phi}{dt}$. This is directly analogous to kinematics, where velocity is the time derivative of position ($v(t) = \frac{dx}{dt}$). While a constant phase rate yields a static frequency f_0 , the heave-induced sinusoidal phase results in a time-varying frequency $f_i(t)$ that physically oscillates around f_0 :

$$f_i(t) = \frac{1}{2\pi} \frac{d\Phi}{dt} = f_0 + \Delta f \cos(2\pi f_w t) \quad (7)$$

where the peak frequency deviation Δf is defined as:

$$\Delta f = \beta f_w = \frac{2\pi f_0 f_w}{c} A_w \cos \theta_g \quad (8)$$

Spectral Expansion (The Bessel Comb)

This subsection transitions the time-domain phase modulation into the frequency domain to mathematically prove that the vessel's physical heave expands the single tonal into an infinite series of frequency sidebands.

[3]

To analyze the frequency spectrum of the modulated signal, we transition to the frequency domain using the **Jacobi-Anger expansion**.

The time-domain pressure wave $p_{tx}(t)$ as the real part of a complex analytic signal:

$$p_{tx}(t) = P_0 \text{Re} \left\{ e^{i2\pi f_0 t} \cdot e^{i\beta \sin(2\pi f_w t)} \right\}$$

After applying the Jacobi-Anger identity, which expands the exponential of a sine function into an infinite series of Bessel-weighted harmonics:

$$e^{i\beta \sin(\phi)} = \sum_{n=-\infty}^{\infty} J_n(\beta) e^{in\phi}$$

where $\phi = 2\pi f_w t$. Substituting this expansion back into the pressure wave equation:

$$p_{tx}(t) = P_0 \text{Re} \left\{ \sum_{n=-\infty}^{\infty} J_n(\beta) e^{i2\pi(f_0 + n f_w)t} \right\}$$

Taking the real part of each term results in an infinite series of discrete spectral lines (sidebands) spaced at intervals of the ocean wave frequency f_w , weighted by Bessel functions of the first kind J_n :

$$p_{tx}(t) = P_0 \sum_{n=-\infty}^{\infty} J_n(\beta) \cos(2\pi(f_0 + n f_w)t) \quad (9)$$

Based on **Carson's Rule**, the physical coupling of the vessel to the ocean surface theoretically expands the frequency spectrum to a bandwidth of $BW \approx 2f_w(\beta + 1)$.

2.2.3 Channel Interaction with Dynamic Boundary & Multipath

This subsection models the reflection off the ocean surface to show how the moving waves act as a dynamic scatterer, imposing a time-varying phase shift (Doppler) that constantly changes the reflected path.

For a distant, submerged receiver, the channel impulse response $h(\tau, t)$ consists of a direct path and a surface-reflected path.

Because the source is at the surface ($H_{src} \approx 0$) and the range is long ($R \gg H_{rec}$), the geometric path length difference between the direct ray and the surface ray approaches zero ($\Delta R \rightarrow 0$), and the time delay $\Delta\tau$ between them is microscopically small ($\Delta\tau \approx 0$).

The received signal $p_{rx}(t)$ is the coherent superposition of the direct path and the surface-scattered path:

$$p_{rx}(t) = \frac{1}{R_{dir}} p_{tx}(t - \tau_{dir}) + \frac{V(t)}{R_{surf}} p_{tx}(t - \tau_{surf}) \quad (10)$$

Where $V(t)$ is the **The Time-Varying Scattering Coefficient**: [4, 5]. The reflection coefficient is not a static -1 . Instead, the moving wave height $h_s(t)$ imparts an additional time-varying phase shift $\phi_s(t)$ upon the reflected ray:

$$\phi_s(t) = 2k_{zn} h_s(t) = \frac{4\pi f_0}{c} h_s(t) \sin \theta_g \quad (11)$$

Taking the time derivative yields the Doppler frequency shift $\Delta f_d(t)$, which is directly proportional to the vertical velocity of the sea surface $v_z(t) = dh_s/dt$:

$$\Delta f_d(t) = \frac{1}{2\pi} \frac{d\phi_s}{dt} = \left(\frac{2f_0}{c} \sin \theta_g \right) v_z(t) \quad (12)$$

Thus, the scattered path is theoretically defined as:

$$V(t) = -R_0 \exp \left(i \frac{4\pi f_0}{c} h_s(t) \sin \theta_g \right) \quad (13)$$

This formulation is the standard Kirchhoff approximation for rough surface scattering. The negative sign represents the 180° phase inversion at the pressure-release air-water boundary. R_0 is the amplitude loss due to rough-surface scattering (energy lost from the specular direction). The exponential term represents the additional phase shift (Doppler) caused by the extra geometric distance the wave must travel as the surface moves vertically.

2.2.4 The Smeared Received Signal

This subsection combines the modulated source signal and the time-varying channel interaction into a single final equation, proving mathematically why the received signal suffers from continuous, multiplicative amplitude fading and phase distortion.

Because $\tau_{dir} \approx \tau_{surf}$ and $R_{dir} \approx R_{surf}$, we can factor the received pressure field into a single equation that defines the phenomenon.

By substituting the FM transmitted signal $p_{tx}(t)$ and the time-varying reflection coefficient $V(t)$ into the superposition equation, the total received signal becomes:

$$p_{rx}(t) \propto p_{tx}(t - \tau_{dir}) \left[1 - R_0 \exp \left(i \frac{4\pi f_0}{c} h_s(t) \sin \theta_g \right) \right]$$

Theoretical Conclusions of the Equation: This final mathematical expression governs exactly why the signal appears smeared:

1. **The Carrier is Modulated:** The baseline signal p_{tx} is already a Bessel-expanded FM signal due to the vessel's physical heave $z(t)$.
2. **Multiplicative Fading:** The term in the brackets represents the micro-multipath interference. It acts as an **amplitude envelope** multiplying the carrier. Because $h_s(t)$ (the ocean wave height) is a random variable evolving over time, this bracketed term oscillates randomly between constructive addition and destructive cancellation.
3. **Secondary Phase Modulation:** The exponential term inside the brackets adds an additional layer of time-varying phase modulation (Doppler spread), driven by the vertical velocity of the surface waves (dh_s/dt).

The Definition of the Smear: The “smeared signal” of a surface vessel is theoretically defined as an inherently frequency-modulated tonal that is continuously subjected to multiplicative, time-varying amplitude fading and secondary phase scattering by a dynamic boundary. It cannot be mathematically reduced to a stationary sine wave.

2.2.5 Connection to the Doubly Spread Channel

This subsection links the derived geometry and physics directly to standard acoustic channel theory, confirming that this specific environment strictly satisfies the formal criteria for both Delay and Doppler spreading.

Yes, this entire framework is the textbook definition of a **doubly spread acoustic channel**.

A channel is doubly spread when the transmitted signal is smeared in both **Delay** (τ) and **Doppler** (ν). The channel spreading function $\chi(\tau, \nu)$ characterizes this mathematically [1, 2].

In this specific geometry:

1. **Delay Spread (τ):** The extremely low θ_g forces the direct path and the surface-scattered paths to travel nearly identical distances ($R_{slant} \approx R$). The delay difference between these paths is so microscopic that they arrive within the same sampling bin, merging into a single unresolvable impulse arrival that fades constructively and destructively.
2. **Doppler Spread (ν):** Simultaneously, the dynamic velocity of the surface $v_z(t)$ and the physical heave of the source $z(t)$ convolve the signal with continuous frequency shifts, spreading the energy across the ν axis.

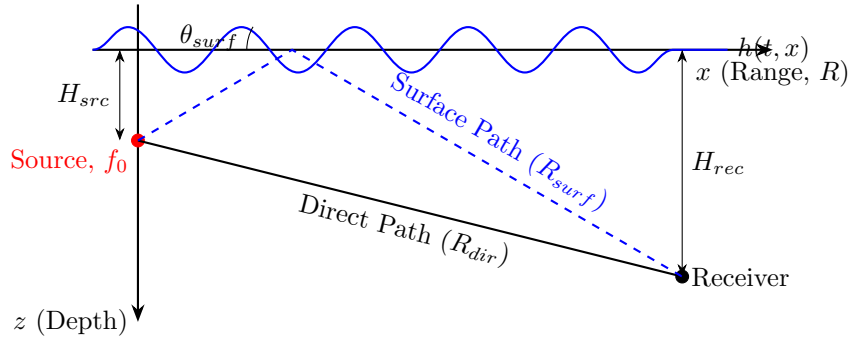
Because the acoustic source is physically bound to the dynamic interface, the signal is instantaneously subjected to both spreading mechanisms simultaneously before it even propagates across the range R .

3 Source Underwater

To highlight the impact of surface coupling, this section contrasts the surface vessel with a deeply submerged source. By isolating the source from the dynamic boundary, we can demonstrate how physical decoupling dramatically stabilizes the acoustic signal.

When inspecting an acoustic source deep in the water column, the acoustic physics changes because the source is decoupled from the ocean's wave spectrum, which results in a more distinct separation of geometric paths and less frequency smearing at the receiver.

3.1 Schematic



3.2 Physical & Mathematical Breakdown

When a vessel is submerged ($H_{src} \gg 0$), it is physically decoupled from the ocean's dynamic boundary. This decoupling mathematically breaks both the kinematic frequency modulation (FM) and the multiplicative micro-multipath fading.

3.2.1 Elimination of Source Kinematics

This subsection proves mathematically that placing the source below the wave base eliminates the physical heave, returning the transmitted signal to a pure, unmodulated tone.

An underwater source operates below the wave base, meaning it is not subjected to the physical heave of the ocean swell.

The vertical heave amplitude approaches zero ($A_w \approx 0$). Consequently, the Modulation Index (β) collapses:

$$\beta = \frac{2\pi f_0}{c} A_w \cos \theta_g \approx 0 \quad (14)$$

Applying $\beta = 0$ to the Jacobi-Anger expansion (the Bessel comb) yields $J_0(0) = 1$ and all higher-order Bessel functions $J_n(0) = 0$. The theoretical infinite bandwidth collapses back into a single singularity. The transmitted signal $p_{tx}(t)$ reverts entirely to a pure, unmodulated monochromatic wave:

$$p_{tx}(t) = P_0 \cos(2\pi f_0 t) \quad (15)$$

meaning the source is emitting a theoretically perfect, infinitely narrow tonal.

3.2.2 Resolution of Macro-Multipath

This subsection shows how the depth of the source geometrically separates the direct path from the surface reflection, creating a macroscopic time delay that breaks the multiplicative interference.

For an underwater vessel, H_{src} is a large, non-zero value. Because of this depth, the geometry creates a significant difference in path lengths:

$$R_{surf} \gg R_{dir} \quad (16)$$

The time delay between the direct path and the surface echo becomes macroscopic ($\Delta\tau = \tau_{surf} - \tau_{dir} \gg 0$).

Because the signals no longer arrive at the same time, they can no longer interfere multiplicatively. The superposition equation is the sum of two separate, additive events in time.

3.2.3 The Decoupled Received Signal

This subsection presents the final received signal equation for an underwater source, proving that the primary acoustic signal arrives completely pristine and is structurally decoupled from surface dynamics.

Substituting our pure transmitted signal (p_{tx}) and our macro-delayed surface reflection into the receiver equation yields the theoretical received signal for an underwater vessel:

$$p_{rx}(t) = \underbrace{\frac{P_0}{R_{dir}} \cos(2\pi f_0(t - \tau_{dir}))}_{\text{Direct Path}} + \underbrace{\frac{P_0 V(t)}{R_{surf}} \cos(2\pi f_0(t - \tau_{surf}))}_{\text{Surface Echo}} \quad (17)$$

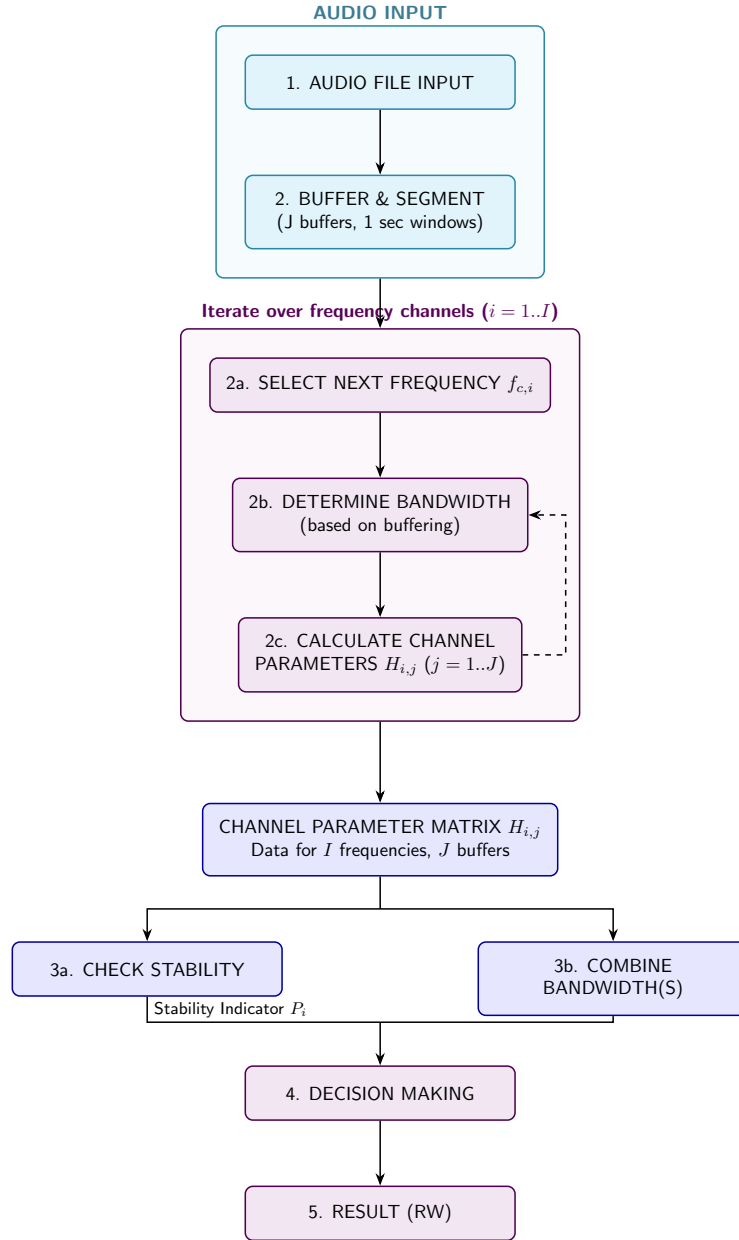
This equation perfectly defines why the main signal is not smeared:

1. **The Direct Path is Pristine:** is a perfect, time-delayed copy of the original engine tonal.
2. **The Echo is Isolated:** the second term contains the time-varying scattering coefficient $V(t)$, which means the Doppler phase modulation from the moving surface does happen to the echo. However, because τ_{surf} is significantly larger than τ_{dir} , it arrives much later.
3. **Additive vs. Multiplicative:** the echo is temporally separated, it adds background reverberation (clutter) to the environment, but it does not multiplicatively modulate or fade the primary direct-path tonal.

The geometry guarantees a pure, stable direct path that is mathematically independent of the surface dynamics. The surface smearing still exists in the environment, but it is confined entirely to a delayed, isolated echo, leaving the primary mechanical tonal sharp and un-smeared.

4 Flow Diagram

Transitioning from theoretical physics to practical application, this section outlines the signal processing architecture. The flow diagram visualizes the algorithm designed to process, analyze, and mitigate the acoustic smearing effects established in the previous sections.



References

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